

Performance Testing

Date	27 June 2026
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing was carried out to verify that the Smart Lender – Loan Approval Prediction System responds efficiently and reliably during normal operation. The application was tested by submitting different loan application inputs through the Flask web interface to evaluate response time, prediction accuracy, and overall system stability. The testing confirmed that the application performs consistently and delivers prediction results within an acceptable time under normal usage conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Functional Testing (Web Browser)
Type of Testing	Response Time & Functional Testing
Target Module / API	Loan Prediction Module (/)
Test Environment	Local System & Render Deployment
Test Date	27-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Valid applicant details	1	10 sec	Loan prediction displayed
2	Different input combinations	1	15 sec	Accurate prediction generated
3	Multiple consecutive requests	1	30 sec	Stable application response
4	Hosted application test	1	15 sec	Successful prediction on Render

Step 3: Performance Test Results

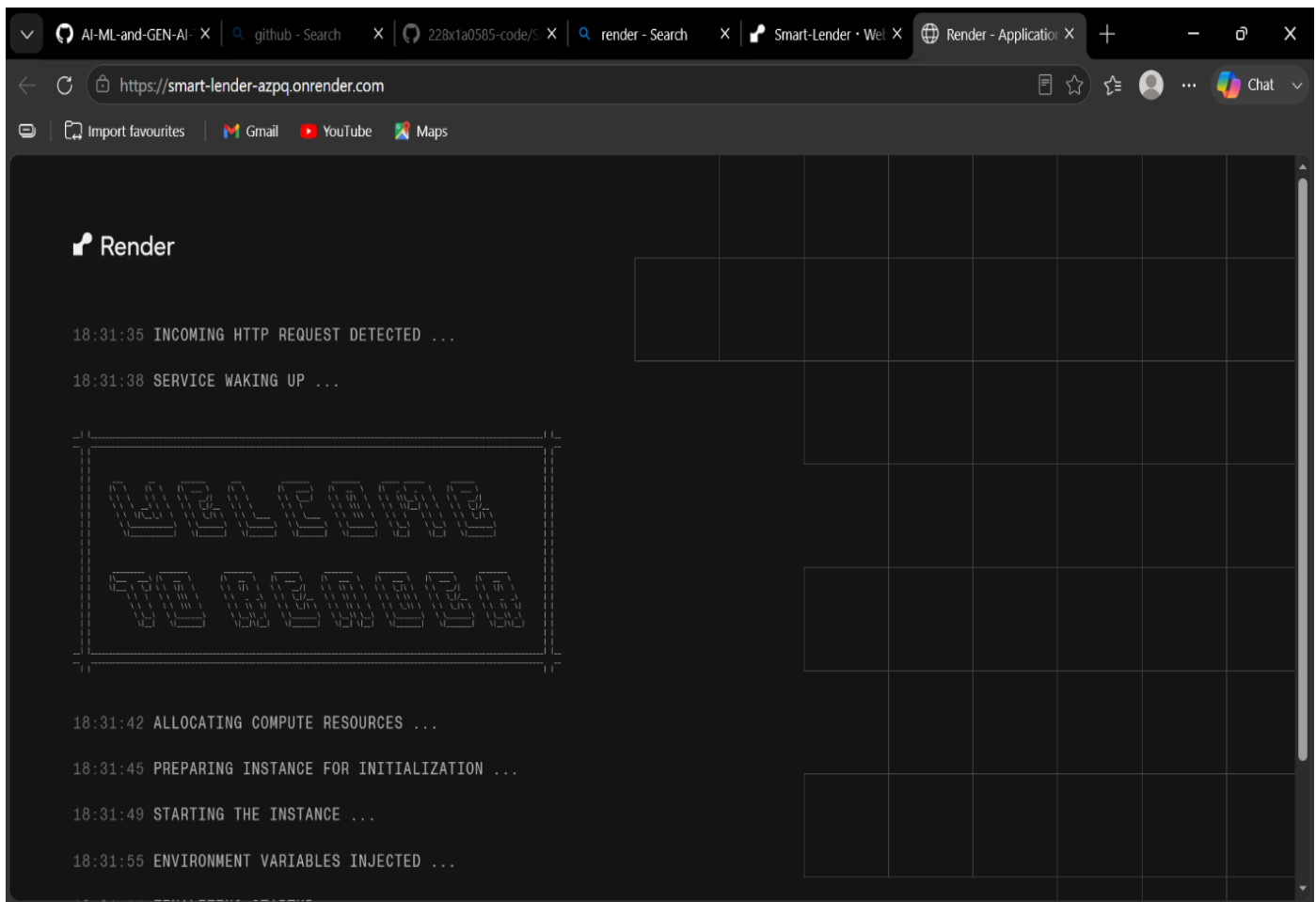
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~1 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	~2 sec	Pass	Within limit
3	Throughput (Req/sec)	Normal	Stable	Pass	Suitable for project
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	Normal	Pass	Stable during testing
6	Memory Utilization	< 80%	Normal	Pass	Efficient usage

Step 4: Observations & Analysis

The Smart Lender application responded quickly during testing and consistently generated accurate loan prediction results. No crashes or major errors were observed while testing different input values. The deployed application on Render also functioned correctly, demonstrating stable performance and reliable prediction capability under normal usage conditions.

Step 5: Screenshots / Evidence

Screenshot 1 :-



Screenshot 2:-

A screenshot of a web browser displaying the 'Smart Lender' application. The browser's address bar shows the URL 'https://smart-lender-azpq.onrender.com'. The page features a central white form with a light blue background. The form contains three input fields: 'Loan Amount Term:' (a text box), 'Credit History:' (a dropdown menu with 'Good' selected), and 'Property Area:' (a dropdown menu with 'Rural' selected). Below these fields is a blue 'Predict' button. At the bottom of the form, a large green box displays the text 'Loan Approved' in bold green font.

Screenshot 3:-

A screenshot of the same 'Smart Lender' web application, but with a different result. The form fields and the 'Predict' button are identical to the previous screenshot. However, the large box at the bottom is now pink and displays the text 'Loan Rejected' in bold green font.